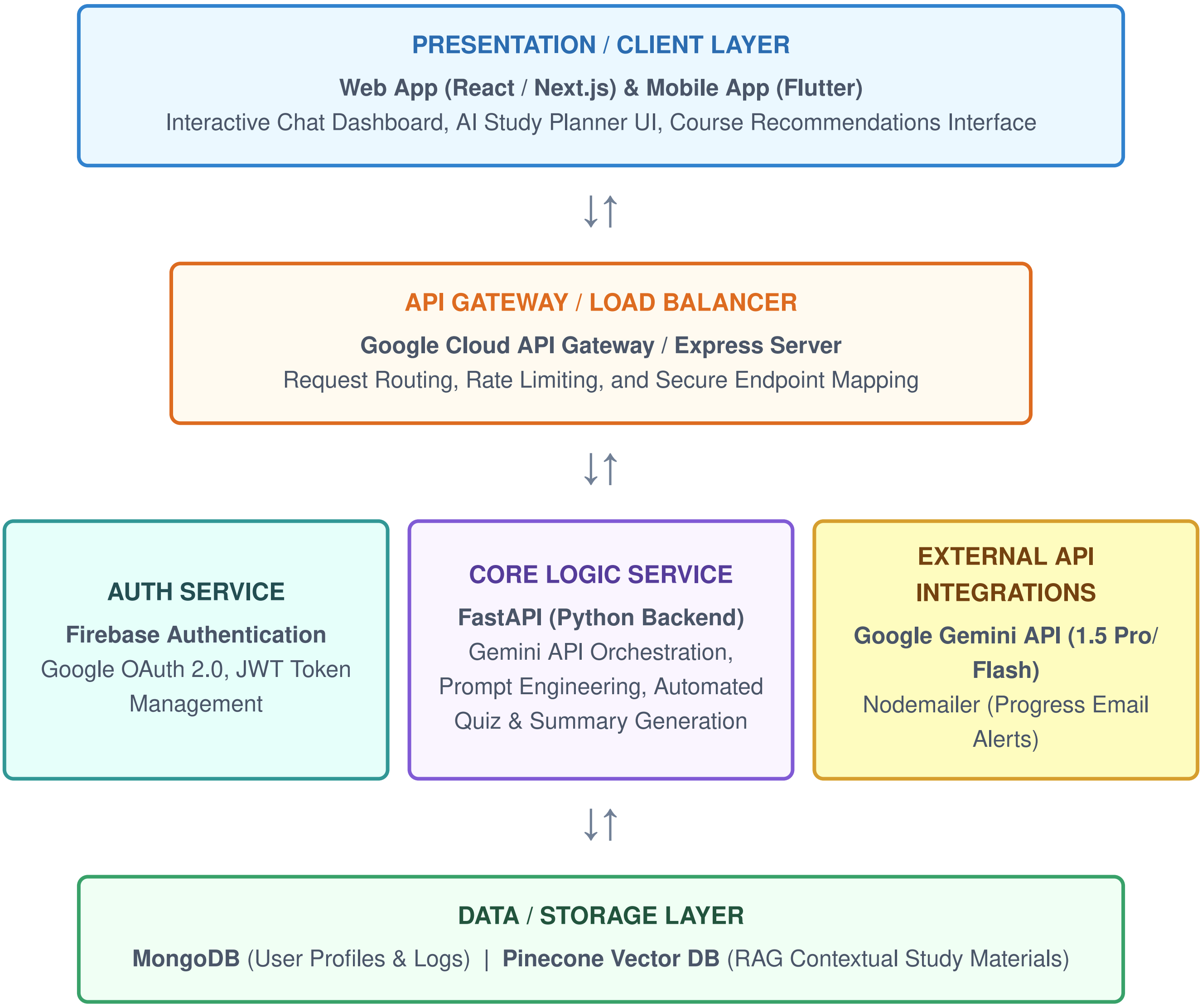
Date	07/07/26
Team ID	XXXXXX
Project Name	EduGenie: Google Gemini-Powered Learning Assistant
Maximum Marks	5 Marks

SOLUTION ARCHITECTURE DIAGRAM



COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides the primary interface for students and educators. Handles interactive AI chat interactions, renders customized multi-modal learning paths, displays dashboards for student analytics, and serves dynamic quiz modules.	React.js, Next.js, Flutter, Tailwind CSS
[API Gateway]	Acts as the single entry point for all client applications. Manages traffic allocation, handles token validation checks, protects internal services against DDoS via rate limiting, and forwards incoming requests safely to underlying microservices.	Google Cloud API Gateway, Node.js / Express Server
[Microservices]	**Auth Service:** Facilitates secure sign-in, session keeping, and RBAC rules. **Core Logic Service:** Processes text/document context uploads, applies precise prompt templates to coordinate with Google Gemini LLMs, executes Retrieval-Augmented Generation (RAG) loops, and compiles auto-generated performance matrices.	FastAPI (Python), Firebase Auth, Google Gemini API SDK
[Database]	stores transactional user metadata, session details, generated quiz scores, and history tracks. Concurrently, a localized vector vector store indexing parses academic textbooks and reference notes to furnish relevant contextual context for precise prompt injections.	MongoDB, Pinecone Vector DB, AWS S3 (for static document assets)